DURATION : 180 Minutes

DATE : 16/02/2025

M.MARKS : 180

ANSWER KEY

PHYSICS

1. (5)
2. (7)
3. (8)
4. (2)
5. (3)
6. (3)
7. (2)
8. (7)
9. (A, C, D)
10. (B, D)
11. (A, B, C)
12. (A, B)
13. (C, D)
14. (A, D)
15. (C)
16. (C)
17. (D)
18. (D)

CHEMISTRY

19. (6)
20. (0)
21. (6)
22. (5)
23. (4)
24. (5)
25. (8)
26. (3)
27. (A, D)
28. (A, B, C, D)
29. (B, C)
30. (A, D)
31. (B, C)
32. (B, C, D)
33. (A)
34. (B)
35. (A)
36. (D)

MATHEMATICS

37. (1)
38. (3)
39. (3)
40. (2)
41. (1)
42. (4)
43. (1)
44. (6)
45. (B, C)
46. (A, C)
47. (B, C, D)
48. (A, B, D)
49. (A, D)
50. (A, C)
51. (D)
52. (B)
53. (A)
54. (D)

PART-I (PHYSICS)

1. **(5)**

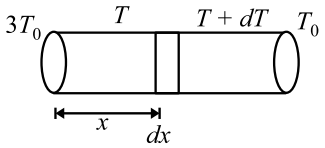
In Steady state

$$\frac{\Delta Q}{\Delta t} = -kA\frac{dT}{dx}$$

$$\frac{\Delta Q}{\Delta t} = -\alpha TA\frac{dT}{dx}$$

$$\frac{\Delta Q}{\Delta t}\int_0^t dx$$

$$= -\alpha TA\int_{3T_0}^{T_0} T dT$$



$$\frac{\Delta Q}{\Delta t}\ell = 4\alpha AT_0^2 \qquad \ldots(i)$$

Similarly $\frac{\Delta Q}{\Delta t}\frac{\ell}{2} = \alpha A\frac{\left(9T_0^2 - T^2\right)}{2}$ …(ii)

Divide (ii) by (i)

$9T_0^2 - T^2 = 4T_0^2$

$T^2 = 5T_0^2$

$T^2 = \sqrt{5}T_0$

2. **(7)**

$$E_x = \frac{qd}{2\pi\varepsilon_0\left(r^2+d^2\right)^{\frac{3}{2}}}$$

Therefore, the electric flux through this Gaussian surface (obtained by integrating over circular shells of perimeter 2πr and thickness *dr*) is

$$\Phi = \int_0^R \frac{qd}{2\pi\varepsilon_0\left(r^2+d^2\right)^{\frac{3}{2}}}2\pi r dr$$

$$= \frac{qd}{\varepsilon_0}\left(\frac{1}{d} - \frac{1}{\sqrt{R^2+d^2}}\right)$$

Now, the fraction captured in a cone with half-angle α, out of the total electric flux emitted by q (which is $\frac{q}{\varepsilon_0}$ by Gauss' law), as $\sin^2\frac{\alpha}{2}$.

Therefore, the above flux must be equal to $\Phi = \frac{q}{\varepsilon_0}\cdot\sin^2\frac{\alpha}{2}$. Solving for $R, R = d\sqrt{\sec^4\frac{\alpha}{2}-1}$

3. **(8)**

$Z_1 = R_1 + X_L i = 8 + 6i$

$Z_2 = R_2 - X_c i = 3 - 4i$

Now, $\frac{1}{Z'} = \frac{1}{Z_1} + \frac{1}{Z_2}$

$$\frac{1}{Z'} = \frac{1}{8+6i} + \frac{1}{3-4i} = \frac{8-6i}{100} + \frac{3+4i}{100} = \frac{20+10i}{100}$$

$$\frac{1}{Z'} = \frac{(2+i)}{10} \Rightarrow \frac{10}{(2+i)}$$

$Z' = 2(2-i)$

$Z' = (4-2i)$ ...(1)

$Z_3 = R_3 + X_L i = (2+10i)$ …(2)

$Z = Z' + Z_3 = 6 + 8i$

$|Z| = \sqrt{(6)^2+(8)^2} = 10\ \Omega$

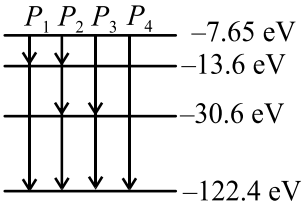
$I = \frac{80}{|Z|} = \frac{80}{10} = 8\ \text{A}$

4. **(2)**

Energy of nth state would be

$$E_n = \frac{-13.6(3)^2}{n^2}eV$$

⇒ Energy diagram would be:



Only possibility P_1 and P_4 would give 1 photon each to eject an electron.

⇒ Number of photons $= \frac{2}{4}\times10^8 = 0.5\times10^8$

⇒ $x = 0.5$

⇒ $\frac{1}{x} = 2$

5. **(3)**

$$g = \frac{GM}{R^2} \qquad \ldots(i)$$

and $v_e = \sqrt{2gR}$ …(ii)

Suppose the velocity of the body at a distance from the earth's center is *v*. Then from conservation of energy

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = \frac{1}{2}mv_e{}^2 - \frac{GMm}{R}$$

⇒ $v = \text{R}\sqrt{\frac{2g}{r}}$

⇒ $\frac{dr}{dt} = \text{R}\sqrt{\frac{2g}{r}}$

$$\Rightarrow \int_0^t dt = \frac{1}{R\sqrt{2g}}\int_R^{R+h}\sqrt{r}\,dr$$

$$\Rightarrow t = \frac{1}{3}\sqrt{\frac{2R}{g}}\left[\left(1+\frac{h}{R}\right)^{3/2}-1\right]$$

6. (3)

Consider a coaxial cylindrical shell of radius x and thickness dx. All such type of cylindrical shells will be in parallel combination. Conductivity at the distance x from the axis

$$\sigma = \sigma_1 + \left(\frac{\sigma_2-\sigma_1}{a}\right)x$$

Therefore, conductance of the cylindrical shell will be $dG = \frac{\sigma(2\pi x dx)}{l}$

$$\Rightarrow G = \int_{x=0}^{x=a}\left[\sigma_1+\left(\frac{\sigma_2-\sigma_1}{a}\right)x\right]\frac{2\pi x}{l}dx$$

$$\Rightarrow G = \frac{\pi a^2(\sigma_1+2\sigma_2)}{3l}$$

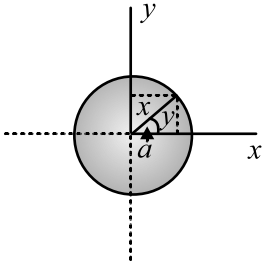
Now, resistance of the conductor $R = \frac{1}{G}$

$$\Rightarrow R = \frac{3l}{\pi a^2(\sigma_1+2\sigma_2)}$$

7. (2)

Consider a current element on the circular ring at an angle θ with the positive x-axis as shown in the figure.

The current induced in the ring, $i = \frac{\pi a^2}{R}\left(\frac{dB_z}{dt}\right)$



$$\Rightarrow i = \frac{\pi a^2\gamma}{R} \quad \ldots(i)$$

Now, $dF = (iad\theta)(\alpha x\cos\theta) + (iad\theta)(\beta y\sin\theta)$

$$\therefore F = 4i\alpha a^2\int_0^{\pi/2}\cos^2\theta d\theta + 4i\beta a^2\int_0^{\pi/2}\sin^2\theta d\theta$$

$$\Rightarrow F = \pi i a^2(\alpha+\beta)$$

$$\Rightarrow F = \frac{\pi^2 a^4(\alpha+\beta)\gamma}{R}[\text{using equation(i)}]$$

8. (7)

$$OC - OA = 2ae$$

$$OB - OA = \left(\sqrt{b^2+(ae)^2}\right) - a(1-e)$$

$$= a - a(1-e) = ae\left(\because e = \sqrt{1-\frac{b^2}{a^2}}\right)$$

From conservation of energy, for the points A and C, $\frac{1}{2}mv_o^2 = \frac{1}{2}k(2ae)^2$ …(i)

For the points *A* and *B*,

$$\frac{1}{2}mv_o^2 + mgb = \frac{1}{2}mv^2 + \frac{1}{2}k(ae)^2 \ldots.(ii)$$

From equations (i) and (ii), we get

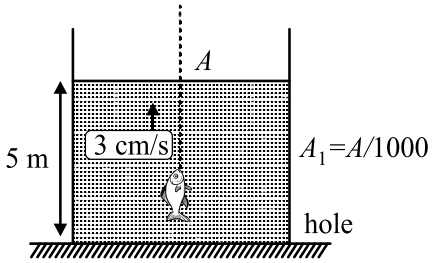
$$v = \sqrt{\frac{3}{4}v_0^2 + 2gb}$$

9. (A, C, D)

(A) Velocity of efflux $v = \sqrt{2g\times 5} = 10$ m/s

Velocity of surface $= \frac{10\times A/1000}{A} = 1$ cm/s

(C) Velocity of fish wrt surface in air $= \frac{3+1}{\mu} = 3$ cm/s



Velocity as viewed directly by the observer =3–1=2 cm/s

(D) $v_i + 1 = \frac{3}{4}(-3+1) = -1.5$

v_i = –2.5 cm/sec

10. (B, D)

Since the wedge is weightless, the sum of the forces acting on it is zero. This means that the horizontal component of the reaction of the body's support on the wedge compensates for the force F_2 Nsin α = *F*.

The support reaction and the force of gravity act on the body.

Newton's second law for a body, vertical direction: $ma_y = N\cdot\cos\alpha - mg$

Newton's second law for a body horizontal direction: $ma_x = N\cdot\sin\alpha$. Acceleration of a body

$$a = \sqrt{a_x^2 + a_y^2}$$

$$= \sqrt{\left(\frac{F}{m\cdot\sin\alpha}\right)^2 + g^2 - 2\frac{F}{m\cdot\sin\alpha}g\cos\alpha}$$

the angle with the horizon is found through the expression.

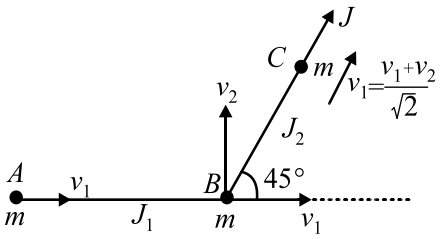
$$\tan\phi = \frac{a_y}{a_x} = \frac{F\cos\alpha - mg\sin\alpha}{F\sin\alpha}$$

$$a = \sqrt{\left(\frac{F}{m\sin\alpha}\right)^2 + g^2 - 2\frac{F}{m\sin\alpha}g\cos\alpha}$$

$$\phi = \tan^{-1}\left(\frac{F\cdot\cos\alpha - mg\cdot\sin\alpha}{F\cdot\sin\alpha}\right)$$

11. (A, B, C)

$$\frac{J_2}{\sqrt{2}} = 2mv_1 \quad \frac{J_2}{\sqrt{2}} = mv_2 \quad \Rightarrow v_2 = 2v_1$$



$$J - J_2 = \frac{m}{\sqrt{2}}(v_1 + v_2)$$

$$J - 2\sqrt{2}mv_1 = \frac{3mv_1}{\sqrt{3}}$$

$$\Rightarrow J = \frac{7mv}{\sqrt{2}} \Rightarrow v_1 = \frac{\sqrt{2}J}{7m}$$

$$v_2 = \frac{2\sqrt{2}J}{7m} \quad v_A = v_1 = \frac{\sqrt{2}J}{7m} \quad v_B = \frac{\sqrt{10}J}{7m}$$

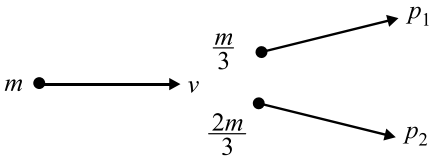
$$v_c = \frac{v_1 + v_2}{\sqrt{2}} = \frac{3J}{7m}$$

12. (A, B)

$$p_1^2 + p_2^2 + 2p_1p_2\cos 60^0 = p^2$$

$$\Rightarrow p_1^2 + p_2^2 + p_1p_2 = p^2 \quad \ldots(i)$$

$$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1} = \frac{2}{1} \Rightarrow p_1 = 2p_2 \quad \ldots(ii)$$



From equation (i) & (ii)

$$p_2 = \frac{p}{\sqrt{7}} = \frac{mv}{\sqrt{7}} \text{ and } p_1 = \frac{2p}{\sqrt{7}} = \frac{2mv}{\sqrt{7}}$$

$$v_1 = \frac{p_1}{m/3} = \frac{6v}{\sqrt{7}} \text{ and } v_2 = \frac{p_2}{2m/3} = \frac{3v}{2\sqrt{7}}$$

Increase in kinetic energy of the system due to explosion

$$K = K_1 + K_2 = \frac{1}{2}\times\frac{m}{3}\times\left(\frac{6v}{\sqrt{7}}\right)^2 + \frac{1}{2}\times\frac{2m}{3}\times\left(\frac{3v}{2\sqrt{7}}\right)^2$$

$$\Rightarrow \quad K = \frac{27}{28}mv^2$$

13. (C, D)

In the right part using adiabatic equation

$300 \times 10^{2/3} = (300 + 2)\, V_2^{2/3}$

$\Rightarrow$ $V_2 = 9.9$ liter)

Pressure is same in left part also temperature in left part T_2' Volume in left part V_2' =20–9.9=10.1 liter

$$T_2' = \left(\frac{10.1\times 302}{9.9}\right) = 308.10$$

By solving $T_2' = 308.10$

$$\therefore \Delta T = 308.10 - 300 = 8.10$$

$$Q = nC_v\Delta T_1 + nC_v\Delta T_2$$

$$= nC_v(\Delta T_1 + \Delta T_2)$$

$$= \frac{P_1V_1}{RT_1}\times\frac{3}{2}R(\Delta T_1 + \Delta T_2) = \frac{3}{2}\frac{P_1V_1}{T_1}(\Delta T_1 + T_2)$$

$$= \frac{3}{2}\times\frac{10^5\times\left(10\times10^{-3}\right)}{300}(2 + 8.10) = 50.50\,\text{J}$$

14. (A, D)

Power given = power loss

$$Fv_T = I^2\left(1+\frac{4}{4}\right) \qquad \ldots(i)$$

$$I = \frac{Bv_T(2r)}{\left(1+\frac{4}{4}\right)} \qquad \ldots(ii)$$

By solving v_T = 64 m/s and I = 8 A

15. (C)

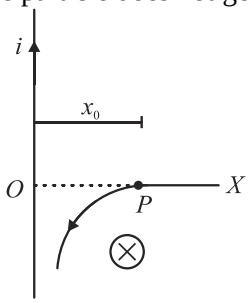
$$\Delta x = \frac{\lambda}{2}(SS_1 + S_1O) - (SO)$$

$$= \frac{\lambda}{2} \quad 2\sqrt{D^2 + d^2} - 2D = \frac{\lambda}{2}2D\left[1 + \frac{d^2}{2D^2} - 1\right]$$

$$= \frac{\lambda}{2}d = \sqrt{\frac{\lambda D}{2}}$$

16. (C)

Let the particle be initially at P (figure). Take the wire as the y-axis and the foot of perpendicular from P to the wire as the origin. Take the line OP as the x-axis. We have, $OP = x_0$. The magnetic field B at any point to the right of the wire is along the negative z-axis. The magnetic force on the particle is, therefore, in the $x - y$ plane. As there is no initial velocity along the z-axis, the motion will be in the $x - y$ plane. Also, its speed remains unchanged. As the magnetic field is not uniform, the particle does not go along a circle.



The force at time t is

$$\vec{F} = q\vec{v}\times\vec{B}$$

$$= q\left(\vec{i}v_x + \vec{j}_y\right)\times\left(-\frac{\mu_0 l}{2\pi t}\vec{k}\right)$$

$$= jqv_x\frac{\mu_0 t}{2\pi x} - iqv_y\frac{\mu_0 d}{2\pi x}$$

Thus $a_x = \frac{F_x}{m} = \frac{-\mu_0 qi}{2\pi m}\frac{v_y}{x} = -\lambda\frac{v_y}{x}$

Where $\lambda = \frac{\mu_0 qi}{2\pi m}$

Also, $a_x = \frac{dv_x}{dt} = \frac{dv_x}{dx}\frac{dx}{dt} = \frac{v_x dv_x}{dx}$

As $v_x^2 + v_y^2 = v^2$, $2v_x dv_x + 2v_y dv_y = 0$

Giving $v_x dv_x = -v_y dv_y$.

From (i), (ii) and (iii)

$\frac{v_y dv_y}{dx} = \frac{\lambda v_y}{x}$ or, $\frac{dx}{x} = \frac{dv_y}{\lambda}$

Initially $t = x_0$ and $v_y = 0$.

At minimum separation from the wire, $v_x = 0$

so that $v_y = -v$.

Thus $\int_{x_0}^{x} \frac{dx}{x} = \int_0^{-v} \frac{dv_y}{\lambda}$

Or, $\ln \frac{x}{x_0} = -\frac{v}{\lambda}$

Or, $x = x_0 e^{-w\lambda} = x_0 e^{-\frac{2\pi n v}{\mu_0 q y^i}}$

17. (D)

$\frac{\mu_0}{4\pi}\left[\frac{2.(2.5)}{5} + \frac{2l}{2}\right] \cdot 1.6\times10^{-19} \times 2\times10^5$

$= 1.6\times10^{-20}$

Solving $I = 4$ A

18. (D)

If the new a adius of the mercury tablet is *R*', then

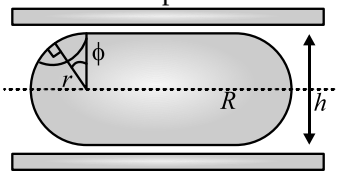
$\left(\pi R^2\right)h = \pi\left(R'\right)^2 \frac{h}{n} \Rightarrow \mathbf{R}' = \mathbf{R}\sqrt{n}$ …(i)

From the geometry

$r = \frac{h}{2\cos\phi}$ …(i)

$\theta = 180 - \phi$ …(ii)

Initial excess pressure inside drop



Initial excess pressure inside drop [using eq. (ii)]

Initial force $p = \frac{T}{r} = \frac{2T\cos\phi}{h}$

$F = p\left(\pi R^2\right) = \frac{2\pi R^2 T\cos\phi}{h}$

Similary final force $F' = \frac{2\pi\left(R'\right)^2 T\cos\phi}{h/n}$

Now, $F' - F = mg$

$\Rightarrow \frac{2\pi\left(R'\right)^2 T\cos\phi}{\left(\frac{h}{n}\right)} - \frac{2\pi R^2 T\cos\phi}{h} = mg$

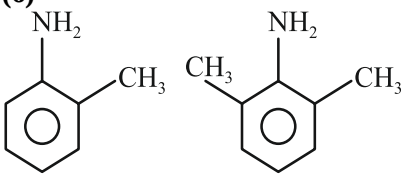
$\cos\phi = \frac{mgh}{2\pi R^2 T\left(n^2-1\right)}$ [using eq.(ii)]

Hence, from equation (iii)

$\theta = \pi - \cos^{-1}\left(\frac{mgh}{2\pi R^2 T\left(n^2-1\right)}\right)$

PART-II (CHEMISTRY)

19. (6)



are less basic than aniline

20. (0)

Element is vanadium.

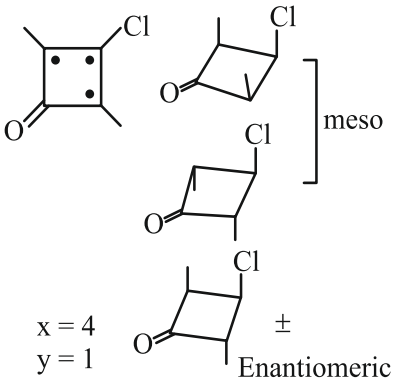
x = 5, y = 5

21. (6)

* Possible Number of geometrical isomers = 6
* All G.I are optically active so total stereoisomers = 12

22. (5)

3-chiral carbon (including one pseudo chiral carbon)



Enantiomeric pair

23. (4)

	P(g)	→	2Q(g)
t = 0	P_0		0
t = 9.6	$P_0 - x$		2x

$\because P_0 - x = 2x$

$\Rightarrow x = \frac{P_0}{3}$

$$\because kt = \ln\left(\frac{P_0}{P-x}\right)$$

$$\Rightarrow t = \frac{t_{1/2}}{\log 2}\log\left(\frac{P_0}{P_0 - \frac{P_0}{3}}\right)$$

$$\Rightarrow 9.6 = \frac{t_{1/2}}{\log 2}\log\frac{3}{2}$$

$$t_{1/2} = \frac{9.6\times\log 2}{\log\frac{3}{2}} \approx 16 \Rightarrow x = 16$$

24. (5)

(i, ii, iii, vi, viii)

The compounds which are more acidic than carbonic acid are soluble in aqueous solution of $NaHCO_3$.

25. (8)

100 ml water contain = 0.192 mg

1000 ml water contain =1.92 mg

Milli mole of $Mg^{+2} = \frac{1.92}{24} = 0.08$

Hence milli mole of $CaCO_3$ = 0.08

Hence mass of $CaCO_3$ with 1L = 0.08 × 100 = 8 mg

Hardness will be = 8 ppm.

26. (3)

$$60 = \frac{hc}{\lambda_1}\left(\frac{n_1}{t_1}\right)$$

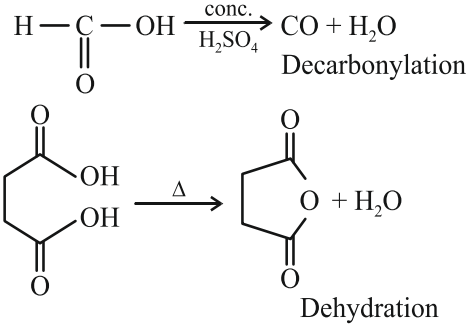
$$80 = \frac{hc}{\lambda_2}\left(\frac{n_2}{t_2}\right)$$

$$\lambda_1 = \frac{1}{R_H}$$

$$\lambda_2 = \frac{1}{4R_H}$$

$$\frac{(n_1/t_1)}{n_2/t_2} = 3$$

27. (A, D)



28. (A, B, C, D)

Theory based.

29. (B, C)

W show +3 oxidation state and X show +2 oxidation state. Y = ns^2np^4, Y show –2 oxidation state Z = $ns^2\ np^5$, Z show –1 oxidation state Formed compounds :

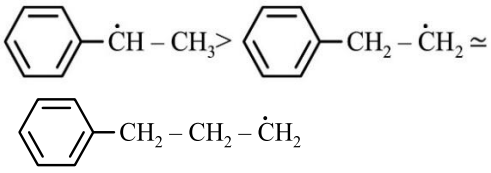
(A) $W^{+3}Y^{-2} = W_2Y_3$

(D) $X^{+2}Z^{-1} = XZ_2$

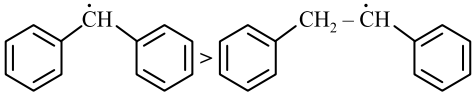
30. (A, D)

The reactivity of compound towards photolytic chlorination is decided by the stability of free radical formed on homolytic cleavage of C–H bond. Correct stability order is

(A) $CH_2 = CH - \dot{C}H_2 > CH_3 - \dot{C}H_2 > \dot{C}H_3$

(B) 

(C) $CH_3CH_2CH_2 - \dot{C}H_2 < CH_3CH_2 - \dot{C}H - CH_3$

(D) 

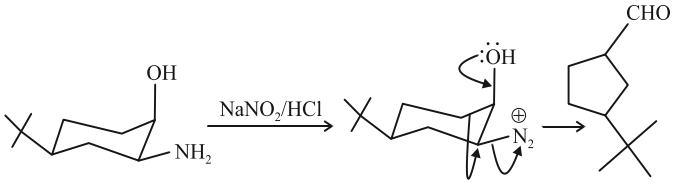
31. (B, C)

 gives test for unsaturation with Br_2 / CCl_4, and Addition reaction with HBr / CCl_4

32. (B, C, D)

Theory based

33. (A)



34. (B)

La^{3+} has $4f^\circ$ and Lu^{+3} has $4f^{14}$ configurations.

Both of them are not expected to show colour in their compounds

35. (A)

Balanced reaction is $2Fe_2O_3 + 3C \rightarrow 4\ Fe + 3CO_2$

Number of moles of Fe_2O_3

$$= \left(\frac{120\times 1000}{(2\times 56 + 48)}\right)\times\frac{90}{100}$$

Mass of 80% pure iron produced

$$= \frac{120 \times 1000 \times 0.9}{(2 \times 56 + 48)} \times \frac{2 \times 56}{0.8}$$

= 94500 gram or 94.5 kg

36. (D)

(A) As K_{SP} of hydroxide of Al^{3+}, Fe^{3+} & Cr^{3-} are low and NH_4Cl suppresses the ionisation of NH_4OH.

(B) $(NH_4)_2SO_4 + Ba^{2+} \rightarrow BaSO_4 \downarrow$ (white) + $2\,NH_4^+$

(C) $BaSO_4$ is insoluble in dil. Mineral acid.

PART-III (MATHEMATICS)

37. (1)

$f(-x) = -f(x) = g(x)$

$\therefore f(-x) \cdot g(x) = -(f(x))^2$ or

$$f(1)g(1) = -(f(1))^2 = -\begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 3 & 1 \end{vmatrix}^2 = -4$$

$\Rightarrow \lambda f(1) g(1) = -2$

$\Rightarrow \lambda(-4) = -2$

$\Rightarrow \lambda = \frac{1}{2}$

38. (3)

$\lambda = 2$

$\therefore [x] = 9, y = 21$

39. (3)

$x + y + z + w < 25$

let

$x + y + z + w + a = 25$

such that $a > 0$ …(i)

$x = -1 + t_1$ $\therefore$ $t_1 \geq 0$

$y = 2 + t_2$ $\therefore$ $t_2 \geq 0$

$z = t_3$ $\therefore$ $t_3 \geq 0$

$w = 4 + t_4$ $\therefore$ $t_4 \geq 0$

$a = 1 + t_5$ $\therefore$ $t_5 \geq 0$

Put in equation (i)

$(-1 + t_1) + (2 + t_2) + t_3 + (4 + t_4) + (1 + t_5) = 25$

$t_1 + t_2 + t_3 + t_4 + t_5 = 19$

The number of non-negative solution

$= {}^{19+5-1}C_{19} = {}^{23}C_{19} = 8855$

40. (2)

$$\frac{1}{2}\left(\frac{1}{\cos 80^\circ} - \frac{\sin 60^\circ}{\cos 60^\circ}\frac{1}{\sin 80^\circ}\right)$$

$$= \frac{1}{2}\left(\frac{\sin 20^\circ}{\cos 80^\circ \cos 60^\circ \sin 80^\circ}\right)$$

$$= \frac{1}{2}(4) = 2$$

41. (1)

$f'(x) = 0 \Rightarrow x = \pm 1$

$\alpha = -1, \beta = 1$

$\therefore \quad M = 3, \; m = \frac{1}{3}$

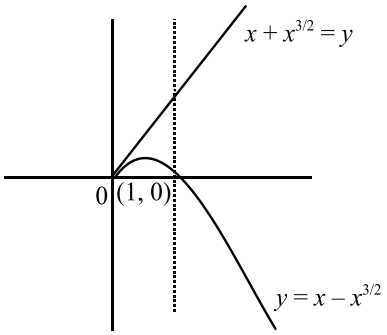
42. (4)

$(y - x)^2 = x^3, x \geq 0$

$\Rightarrow y = x \pm x^{3/2}$

Required area

$$A = \int_0^1 \left(\left(x + x^{3/2}\right) - \left(x - x^{3/2}\right)\right) dx = \frac{4}{5} \text{ sq. unit}$$



43. (1)

$2010^2 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 67^2$

So, total no of divisors

$= (2 + 1) \cdot (2 + 1) \cdot (2 + 1) \cdot (2 + 1)$

$= 3^4 = 81$

Total no divisors that are perfect square = 16

Now, no of ways to select 1 perfect square from two selected distinct divisors

$= {}^{16}C_1 \cdot {}^{65}C_1$

$= 16.65 = 1040$

Prob = $\frac{1040}{{}^{81}C_2} = \frac{1040}{3240} = \frac{26}{81}$

44. (6)

$\vec{P}_1 \cdot \vec{P}_2 = 0 \Rightarrow a + b + c = 0$

$\frac{a}{-2}$	$\frac{b}{1}$	$\frac{c}{1}$		
			$\Rightarrow$	$\frac{3!}{2!} = 3$
–2	0	2	$\Rightarrow$	3! = 6
–1	0	1	$\Rightarrow$	3! = 6
–1	–1	2	$\Rightarrow$	$\frac{3!}{2!} = 3$

Total number of vectors = 18

45. **(B, C)**

$l_1 = \int_0^{512} \frac{\{x^{1/3}\}}{\sqrt[3]{x^2}}\,dx.$

Let $x^{1/3} = t \Rightarrow \frac{dt}{dx} = \frac{1}{3}x^{-2/3}$

$\Rightarrow l_1 = 3\int_0^8 \{t\}\,dt = 3 \times 8 \times \int_0^1 \{t\}\,dt = 24 \times \frac{1}{2} =$ 12

$l_2 = \int_0^2 x^2\{x^3\}\,dx$. Let $x^3 = t \Rightarrow 3x^2\,dx = dt$

$= \frac{1}{3}\int_0^8 \{t\}\,dt = \frac{1}{3} \times 8 \times \int_0^1 \{t\}\,dt = \frac{8}{3} \times \frac{1}{2} = \frac{4}{3}$

46. **(A, C)**

By venn diagram $n(A \cap B) = 2, n(\overline{A \cup B}) = 5$

47. **(B, C, D)**

$y = f(x) = \cos\{[x-3] + \{x - \{x - \{x\}\}\}\}$
$= \cos\{x - \{x - \{x\}\}\} = \cos\{x - \{[x]\}\}$
$= \cos\{x - 0\} = \cos\{x\} = \cos x$

$\because \quad 0 < x < \frac{\pi}{4}$

$\therefore \quad x = \cos^{-1} y$ or $f^{-1}(x) = \cos^{-1} x$

48. **(A, B, D)**

$\left(\frac{1}{x-1}+1\right) - \left(\frac{4}{x-4}+1\right) - \left(\frac{5}{x-5}+1\right) + \left(\frac{8}{x-8}+1\right)$

$= \frac{6x^2 - 27x}{40} \qquad \Rightarrow x = 0$

Also $\frac{2x-9}{(x-1)(x-8)} - \frac{2x-9}{(x-4)(x-5)} = \frac{3}{40}(2x-9)$

Now $2x - 9 = 0$

$\Rightarrow \quad x = \frac{9}{2}$

Also $x = 9$

49. **(A, D)**

Graphically
$\alpha = 0, \beta = 2$

50. **(A, C)**

$h(f(x))$ is continuous $\forall\ x \in R$ and not differentiable at $x = -1, 1, 2$

51. **(D)**

For a, b, c

Mean $\frac{a+b+c}{3}(= \bar{x})$

$b = a + c$

$\Rightarrow \quad \bar{x} = \frac{2b}{3} \quad ....(1)$

S.D. $(a + 2, b + 2, c + 2)$ = S.D. $(a, b, c) = d$

$\Rightarrow \quad d^2 = \frac{a^2+b^2+c^2}{3} - (\bar{x})^2$

$\Rightarrow \quad d^2 = \frac{a^2+b^2+c^2}{3} - \frac{4b^2}{9}$

$\Rightarrow \quad 9d^2 = 3(a^2 + b^2 + c^2) - 4b^2$
$\Rightarrow \quad b^2 = 3(a^2 + c^2) - 9d^2$

52. **(B)**

Coeff of x^{10} in $(1 + x)^{15}(x + 1)^{35}$ for 10 terms gives above sum.

53. **(A)**

$2ax + 2hxy' + 2hy + 2by' = 0$ also
$x(ax + hy) + y(hx + by) = 0$

$y' = \frac{y}{x}$

54. **(D)**

$\left(\hat{i} + k\hat{j} + 2\hat{k}\right) \cdot \left(\hat{i} + \hat{j} + \hat{k}\right) = 0$

$k = -3$

$\log_{\frac{1}{3}} 27 = -3$

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